

1. Administrative & Study Identification

Full Study Title: A Phase 2b, Single-Arm, Open-Label, Multicenter Study of the Safety of SPK-8011QQ in Adults With Severe or Moderately Severe Hemophilia A **Short Title/Acronym:** A Gene Therapy Study of SPK-8011QQ in Adults With Severe or Moderately Severe Hemophilia A **Protocol Number:** X046084 / SPK-8011QQ-201 **NCT Number:** NCT07226206 **Sponsor Name:** Hoffmann-La Roche (Spark Therapeutics) **Investigational Product:** SPK-8011QQ (Gene Therapy Vector) **Phase of Development:** Phase II (Phase 2b) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of a single infusion of the SPK-8011QQ gene therapy in adult males with moderately severe to severe hemophilia A. **Secondary Objectives:** To evaluate the preliminary efficacy, pharmacokinetics, and pharmacodynamics of SPK-8011QQ, focusing on long-term endogenous Factor VIII (FVIII) expression and reduction in bleeding events.

2.2 Background & Rationale To address the unmet medical need in patients with severe or moderately severe hemophilia A (endogenous FVIII:C activity levels $\leq 3\%$) who otherwise require frequent, lifelong Factor VIII replacement therapy. SPK-8011QQ offers a potential one-time, adeno-associated virus (AAV) vector-mediated gene transfer intended to deliver sustained FVIII expression, thereby reducing the dependency on regular exogenous prophylaxis and minimizing recurrent bleeding episodes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 5 evaluable patients **Number of Sites:** Multicenter **Estimated Study Start:** 31-Mar-2026 **Estimated Primary Completion:** 31-Jul-2031

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male ≥ 18 years of age at the time of signing the ICF. 2. Diagnosis of severe or moderately severe hemophilia A, defined as endogenous FVIII:C activity levels $\leq 3\%$

3. Confirmed negative anti-Spark200 antibodies as documented through central laboratory testing, and acceptable hepatobiliary/renal function (creatinine clearance ≥ 30 mL/min/1.73 m²).

4.2 Exclusion Criteria 1. Prior treatment with a vector or gene transfer agent (excluding nucleic acid-based vaccines like COVID-19 vaccines). 2. Currently undergoing antiviral therapy for chronic hepatitis B or chronic hepatitis C, or having an inherited/acquired bleeding disorder other than hemophilia A. 3. Having received an investigational drug within 30 days or having known inherited/acquired thrombophilia.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of SPK-8011Q. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** One-time administration with a 5-year long-term observation and safety follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Exogenous FVIII replacement therapy for the treatment of acute bleeding episodes (if necessary). **Prohibited:** Concurrent investigational drugs, other gene therapies, or hepatotoxic medications during the primary evaluation window.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -45 to -1	Informed Consent, Medical History, Anti-Spark200 antibody testing, FVIII:C Labs, LFTs
Baseline	Day 0	Admission, Single IV Infusion of SPK-8011Q, Vital Signs Monitoring
Interim	Weeks 1-52	Intensive Safety Labs (especially liver transaminases to monitor vector immune response), FVIII activity assessment, Bleeding diary review
End of Study	Year 5 (2031)	Final Vital Signs, Exit Interview, Long-term Efficacy/Safety Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile, determined by the incidence, severity, and relatedness of Adverse Events (AEs) and Serious Adverse Events (SAEs). **Measurement Method:** Clinical observation and central laboratory analysis (e.g., frequent LFT monitoring for AAV-mediated hepatotoxicity).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in endogenous FVIII:C activity levels over time. 2. Annualized Bleeding Rate (ABR) and reduction in the annualized number of exogenous FVIII replacement therapy infusions.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient bleeding diaries, routine HCP interviews, and rigorous laboratory monitoring for vector-induced immune responses (e.g., transaminitis). **Serious Adverse Events (SAEs):** Required reporting to the sponsor and Institutional Review Board (IRB) within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee (IDMC) to review safety signals continuously.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients who receive any portion of the SPK-8011QQ infusion). **Sample Size Justification:** N=5 participants, which is adequately powered for the initial Phase 2b safety and tolerability objectives in this rare disease indication. **Handling of Missing Data:** Descriptive statistics will be utilized without imputation for safety endpoints. Mixed models for repeated measures (MMRM) or LOCF may be utilized for evaluating FVIII expression trajectories.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring post-infusion, shifting to remote monitoring for the long-term follow-up period. **Audit Trail:** Validated Electronic Data Capture (EDC) system including eCRF locking and comprehensive data clarification processes.

11. Study Identifiers & Governance

Primary ID: X046084 **Registry Number:** NCT07226206 **Sponsor/Lead Organization:** Hoffmann-La Roche (Spark Therapeutics) **Study Title:** A Gene Therapy Study of SPK-8011QQ in Adults With Severe or Moderately Severe Hemophilia A **Official Regulatory Title:** A Phase 2b, Single-Arm, Open-Label, Multicenter Study of the Safety of SPK-8011QQ in Adults With Severe or Moderately Severe Hemophilia A

12. Clinical Framework (Hemophilia A Specific)

Primary Condition: Severe or Moderately Severe Hemophilia A
Diagnosis Threshold: Endogenous FVIII:C activity levels $\leq 3\%$

Medical Methodology: Adeno-associated virus (AAV) Vector-Mediated Gene Therapy **Optimization Target:** Sustained endogenous Factor VIII expression and bleeding cessation

13. Study Procedures & Methodology

Management Pathway: Specialized gene therapy administration and immune monitoring clinical pathway **Education Protocol (HCP):** Vector handling, preparation protocols, infusion observation, and post-infusion steroid/immunosuppressant management guidelines **Education Protocol (Patient):** Strict guidelines on recognizing hepatotoxicity symptoms, bleeding monitoring, and recording interventions in the study diary **Quality Audit Mechanism:** Sponsor audits of site dosing, sterile prep protocols, and patient bleeding diary adherence

14. Observation & Follow-Up Schedule

Total Duration: Approx. 260 Follow-up weeks (5 Years) **Onsite Visit Cadence:** Weekly/Bi-weekly during Weeks 1-12; progressively tapering to Q12W and then semi-annual/annual visits. **Remote Monitoring Frequency:** Intermittent tele-health visits during the long-term follow-up phase. **Checklist Review Interval:** AE, bleeding events, and FVIII replacement log reviewed at every contact point.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				